Cambridge (CIE) A Level Maths: Pure 3

Your notes

Further Complex Numbers

Contents

- Exponential Form of Complex Numbers
- Geometry of Complex Multiplication & Division
- Square Roots of a Complex Number using Exponential Form

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Exponential form of complex numbers

You now know how to do lots of operations with complex numbers: add, subtract, multiply, divide, raise to a power and even square root. The last operation to learn is raising the number e to the power of an imaginary number.

What is e to the power of an imaginary number?

- Given an imaginary number (iθ) we can **define exponentiation** as
  - $e^{i\theta} = \cos\theta + i\sin\theta$
  - $e^{i\theta}$ is the **complex number** with **modulus 1** and **argument θ**
- This works with our current rules of exponents
  - $e^0 = e^{0i} = \cos 0 + i\sin 0 = 1$
    - This shows e to the power 0 would still give the answer of 1
  - $e^{i\theta_1} \times e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$
    - This is because when you **multiply complex numbers** you can **add** the **arguments**
    - This shows that when you multiply two powers you can still add the indices
  - $\dfrac{e^{i\theta_1}}{e^{i\theta_2}} = e^{i(\theta_1 - \theta_2)}$
    - This is because when you **divide complex numbers** you can **subtract** the **arguments**
    - This shows that when you divide two powers you can still subtract the indices

What is the exponential form of a complex number?

- Any complex number $z = a + bi$ can be written in **polar form** $z = r(\cos\theta + i\sin\theta)$
  - *r* is the modulus
  - θ is the argument
- Using the definition of $e^{i\theta}$ we can now also write z in **exponential form**
  - $z = re^{i\theta}$

Why do I need to use the exponential form of a complex number?

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- It's just a **shorter** and **quicker** way of expressing complex numbers
- It makes a link between the **exponential function** and **trigonometric functions**
- It makes it easier to remember what happens with the moduli and arguments when multiplying and dividing
- If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ then
  - $z_1 \times z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$
    - You can clearly see that the **moduli have been multiplied** and the **arguments have been added**
  - $\dfrac{z_1}{z_2} = \dfrac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$
    - You can clearly see that the **moduli have been divided** and the **arguments have been subtracted**

What are some common numbers in exponential form?

- As $\cos(2\pi) = 1$ and $\sin(2\pi) = 0$ you can write:
  - $1 = e^{2\pi i}$
- Using the same idea you can write:
  - $1 = e^0 = e^{2\pi i} = e^{4\pi i} = e^{6\pi i} = e^{2k\pi i}$ where *k* is any integer
- As $\cos(\pi) = -1$ and $\sin(\pi) = 0$ you can write:
  - $e^{\pi i} = -1$
  - Or more commonly written as $e^{i\pi} + 1 = 0$
- As $\cos\left(\dfrac{\pi}{2}\right) = 0$ and $\sin\left(\dfrac{\pi}{2}\right) = 1$ you can write:
  - $i = e^{\frac{\pi}{2} i}$

Worked Example

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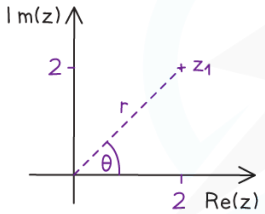
Two complex numbers are given by $z_1 = 2 + 2\text{i}$ and $z_2 = 3e^{\frac{2\pi}{3}\text{i}}$.

(a) Write z_1 in the form $re^{\text{i}\theta}$.

(b) Write z_2 in the form $a + b\text{i}$.

(c) Calculate $z_1 \times z_2$ giving your answer in the form $r(\cos\theta + \text{i}\sin\theta)$.

a)



r IS THE MODULUS

$r = |z_1| = \sqrt{2^2 + 2^2} = 2\sqrt{2} = 2.83$ (3sf)

θ IS THE ARGUMENT

$\theta = \tan^{-1}\left(\frac{2}{2}\right) = \frac{\pi}{4} = 0.785$ (3sf)

$z_1 = 2\sqrt{2}e^{\frac{\pi}{4}\text{i}}$

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b)

USE $e^{\text{i}\theta} = \cos\theta + \text{i}\sin\theta$

$z_2 = 3\left(\cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}\right)$

EVALUATE EACH TERM

$= 3\left(-\frac{1}{2} + \text{i}\frac{\sqrt{3}}{2}\right)$

EXPAND

$z_2 = -\frac{3}{2} + \text{i}\frac{3\sqrt{3}}{2}$

$z_2 = -1.5 + 2.60\text{i}$ (3sf)

c)

USE EXPONENTIAL FORM TO MULTIPLY

$z_1 \times z_2 = 2\sqrt{2}e^{\frac{\pi}{4}\text{i}} \times 3e^{\frac{2\pi}{3}\text{i}}$

ADD INDICES MULTIPLY CONSTANTS AT THE FRONT

$= 2\sqrt{2} \times 3e^{\left(\frac{\pi}{4}\text{i} + \frac{2\pi}{3}\text{i}\right)}$

$= 6\sqrt{2}e^{\frac{11\pi}{12}\text{i}}$

USE $e^{\text{i}\theta} = \cos\theta + \text{i}\sin\theta$

$z_1 \times z_2 = 6\sqrt{2}\left(\cos\frac{11\pi}{12} + \text{i}\sin\frac{11\pi}{12}\right)$

$z_1 \times z_2 = 10.4\,(\cos 2.88 + \text{i}\sin 2.88)$ (3sf)

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Examiner Tips and Tricks

- The powers can be long and contain fractions so make sure you write the expression clearly.
- You don't want to lose marks because the examiner can't read your answer

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Geometry of complex multiplication & division

You now know how conjugation, addition and subtraction affect the geometry of complex numbers on an Argand diagram. Now we can look at the effects of multiplication and division.

How do I sketch complex multiplication and division on an Argand diagram?

- Let z_1 and z_2 be two complex numbers
  - With moduli r_1 and r_2 respectively
  - And arguments θ_1 and θ_2 respectively
- To plot $z_1 \times z_2$ on an Argand diagram
  - The modulus will be $r_1 \times r_2$
  - The argument will be $\theta_1 + \theta_2$
    - Subtract 2π from the argument if it is not in the range
  - To plot $\frac{z_1}{z_2}$ on an Argand diagram
    - The modulus will be $\frac{r_1}{r_2}$
    - The argument will be $\theta_1 - \theta_2$
      - Add 2π to the argument if it is not in the range

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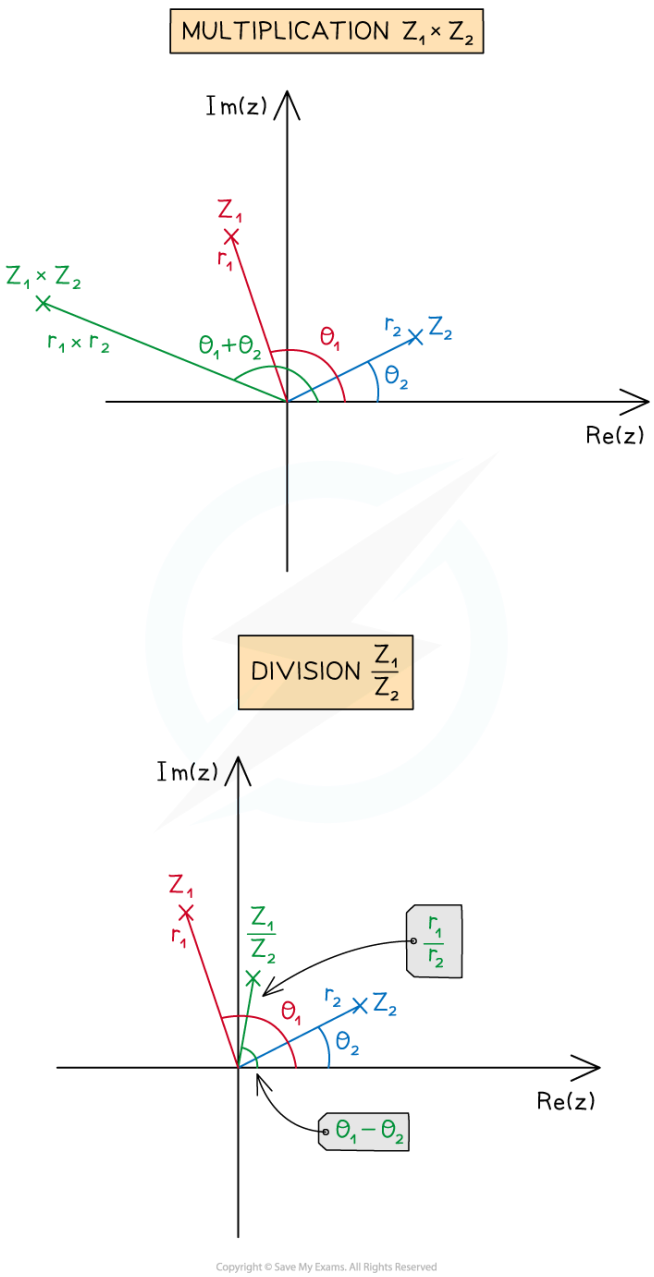

Which geometric transformations represent complex multiplication and division?

- Let *w* be a given complex number with **modulus *r*** and **argument θ**
  - In exponential form $w = re^{i\theta}$
- Let *z* be any complex number represented on an Argand diagram
- **Multiplying *z* by *w*** results in *z* being:
  - **Stretched** from the origin by a **scale factor of *r***
    - If $r > 1$ then *z* will move further away from the origin

Your notes

- If $0 < r < 1$ then z will move closer to the origin
- If $r = 1$ then z will remain the same distance from the origin

- **Rotated anti-clockwise** about the origin by **angle θ**
  - If $\theta < 0$ then the rotation will be clockwise

- **Dividing *z* by *w*** results in z being:
  - **Stretched** from the origin by a **scale factor of** $\frac{1}{r}$
    - If $r > 1$ then z will move closer to the origin
    - If $0 < r < 1$ then z will further away from the origin
    - If $r = 1$ then z will remain the same distance from the origin
  - **Rotated clockwise** about the origin by **angle θ**
    - If $\theta < 0$ then the rotation will be anti-clockwise

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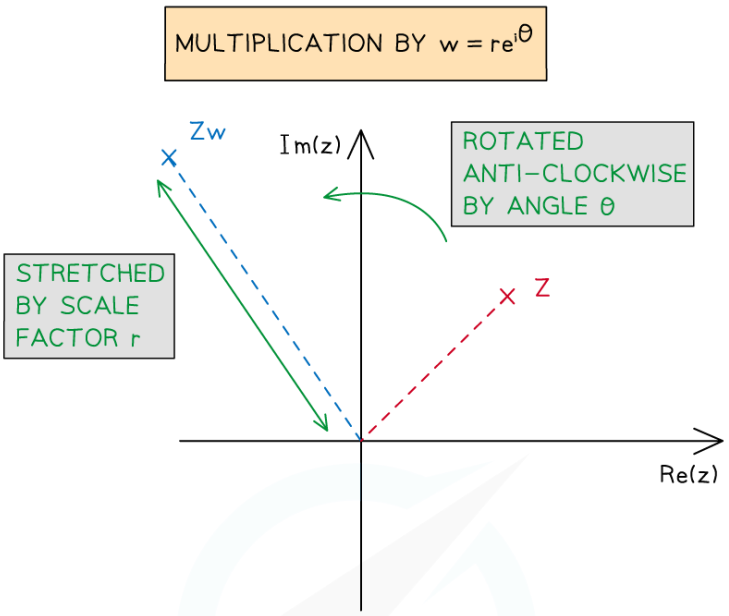
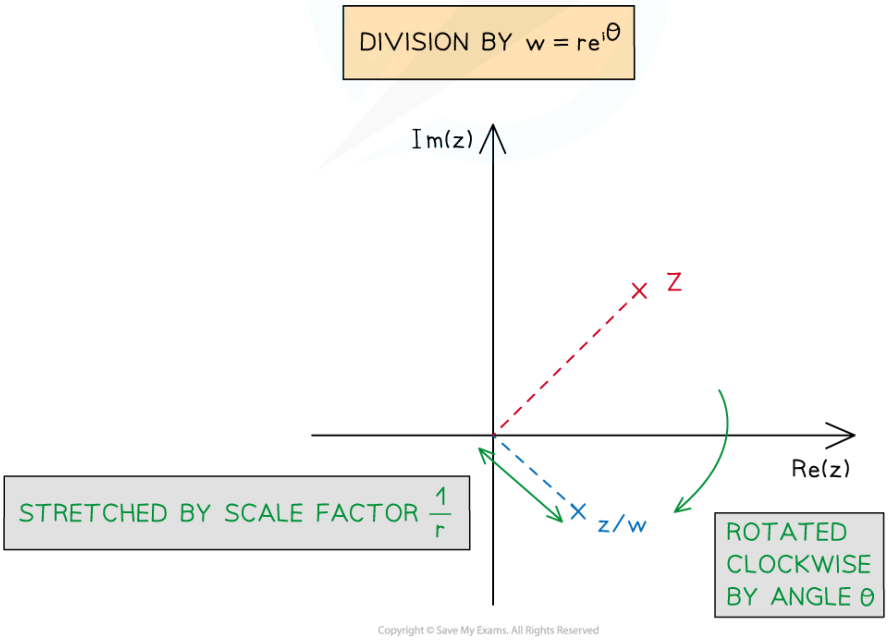




Worked Example

Your notes

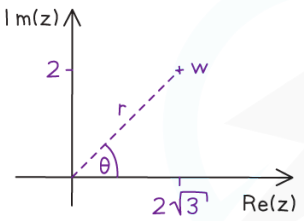
A complex number is given by $w = 2\sqrt{3} + 2i$.

A complex number z is plotted on an Argand diagram. Describe the geometric effects on z when:

(i) z is multiplied by w,

(ii) z is divided by w.

FIND THE MODULUS AND ARGUMENT OF w



$$|w| = \sqrt{(2\sqrt{3})^2 + (2)^2} = 4$$

$$\arg w = \tan^{-1}\left(\frac{2}{2\sqrt{3}}\right) = \frac{\pi}{6}$$

(i) MULTIPLYING BY w

- STRETCHES z FROM THE ORIGIN BY SCALE FACTOR 4 ← $r = |w|$
- ROTATES z ANTI-CLOCKWISE ABOUT THE ORIGIN BY $\frac{\pi}{6}$ ← $\theta = \arg w$

(ii) DIVIDING BY w

- STRETCHES z FROM THE ORIGIN BY SCALE FACTOR $\frac{1}{4}$ ← $\frac{1}{r}$
- ROTATES z CLOCKWISE ABOUT THE ORIGIN BY $\frac{\pi}{6}$

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Examiner Tips and Tricks

- If a complex number is given in Cartesian form, first convert it to polar form or exponential form to find the modulus and argument.

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Square roots of a complex number using exponential form

Previously we looked at how to find the square roots of a complex number in Cartesian form (a+bi). We can also find square roots using polar ($r(\cos\theta + \mathrm{i}\sin\theta)$) and exponential form ($re^{\mathrm{i}\theta}$).

How do I find a square root of a complex number in exponential form?

- Let $w = re^{\mathrm{i}\theta}$ be a square root of $z = ne^{\mathrm{i}\alpha}$
  - $w \times w = z$
  - $(re^{\mathrm{i}\theta})^2 = ne^{\mathrm{i}\alpha}$
- Applying rules of indices:
  - $r^2 e^{2i\theta} = ne^{\mathrm{i}\alpha}$
- Comparing the coefficients of e (moduli) and powers of e (arguments) we can state:
  - $n = r^2$
    - $r = \sqrt{n}$
  - $\alpha = 2\theta$
    - $\theta = \dfrac{\alpha}{2}$
- A square root of $z = ne^{\mathrm{i}\alpha}$ is $w = \sqrt{n}\, e^{\frac{\alpha}{2}\mathrm{i}}$
  - Square root the modulus
  - Halve the argument

How do I find the second square root?

- The second square root is the first one multiplied by -1
  - $w = \sqrt{n}\, e^{\frac{\alpha}{2}\mathrm{i}}$ and $-w = -\sqrt{n}\, e^{\frac{\alpha}{2}\mathrm{i}}$
- We can write the second one in polar or exponential form too
- Adding 2π to the argument of a complex number still gives the same complex number

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- So we could also say that $ne^{i(\alpha+2\pi)} = r^2 e^{2i\theta}$
- Therefore $\alpha + 2\pi = 2\theta$ is another possibility
  - $\theta = \frac{\alpha}{2} + \pi$
- So the two square roots of ($ne^{i\alpha}$) are:
  - $z_1 = \sqrt{n}\ e^{\frac{\alpha}{2}i}$
  - $z_2 = \sqrt{n}\ e^{\left(\frac{\alpha}{2}+\pi\right)i}$
- You should notice that the two square roots are π radians apart from each other
  - This is always true when finding square roots
- And if you were to write them in cartesian form they would be negatives of one another
  - E.g. a+bi and -a-bi
  - This is also always true when finding square roots
- This approach can be extended to find higher order roots (e.g. cube roots) by knowing that the nth roots will be $\frac{2\pi}{n}$ radians apart from each other, however this is beyond the specification of this course

Examiner Tips and Tricks

- The square roots will be negatives of each other when written in cartesian form, and the two square roots will be π radians apart when written in polar form. These two facts can help you find the roots quicker and/or check your answers.
- If your calculator is able to work with complex numbers, you should also square the square-roots you found to check that you get the original number.

Worked Example

Your notes

Find the square roots of $2 + 2i\sqrt{3}$.

$2 + 2i\sqrt{3} = z^2$

EXPRESS THE COMPLEX NUMBER AS z^2, WHERE z REPRESENTS THE SQUARE ROOTS WE ARE TRYING TO FIND

$4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = \left(r\left(\cos\theta + i\sin\theta\right)\right)^2$

EXPRESS BOTH SIDES IN POLAR FORM. THE RIGHT-SIDE IS A GENERAL COMPLEX NUMBER; SO WE CAN JUST USE r FOR ITS MAGNITUDE, AND θ FOR ITS ARGUMENT

$4e^{i\frac{\pi}{3}} = \left(re^{i\theta}\right)^2$

USE THIS TO EXPRESS BOTH SIDES IN EXPONENTIAL FORM (YOU COULD GO STRAIGHT TO THIS STEP IF YOU ARE CONFIDENT)

$4e^{i\frac{\pi}{3}} = r^2e^{2i\theta}$

EXPAND THE RIGHT HAND SIDE USING LAWS OF INDICES: $(ab)^n = a^nb^n$ AND $(a^n)^m = a^{nm}$

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COMPARING THE COEFFICIENTS OF e (THE MODULII):

$4 = r^2$

$\Rightarrow 2 = r$

COMPARING THE POWERS OF e (THE ARGUMENTS):

$\frac{\pi}{3} = 2\theta$

$\Rightarrow \frac{\pi}{6} = \theta$

WE KNOW THAT ADDING 2π TO THE ARGUMENT OF A COMPLEX NUMBER, STILL GIVES THE SAME COMPLEX NUMBER. SO WE COULD ALSO SAY THAT:

$4e^{i\left(\frac{\pi}{3} + 2\pi\right)} = r^2e^{2i\theta}$

$\Rightarrow \frac{\pi}{3} + 2\pi = 2\theta$

$\Rightarrow \frac{7\pi}{6} = \theta$

COMPARING THE ARGUMENTS AGAIN

WE NOW HAVE THE MODULUS, AND TWO DIFFERENT ARGUMENTS FOR OUR SQUARE ROOTS:

$z_1 = 2e^{i\frac{\pi}{6}}$ $\quad z_2 = 2e^{i\frac{7\pi}{6}}$

WE COULD ALSO EXPRESS THEM IN POLAR FORM:

$z_1 = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$ $\quad z_2 = 2\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right)$

NOTICE THAT THE ARGUMENTS ARE π RADIANS APART; THIS IS ALWAYS TRUE FOR SQUARE ROOTS. WE COULD ALSO EXPRESS THEM IN CARTESIAN FORM:

$z_1 = \sqrt{3} + i$ $\quad z_2 = -\sqrt{3} - i$

NOTICE THAT THEY ARE NEGATIVES OF ONE ANOTHER; THIS IS ALWAYS TRUE FOR SQUARE ROOTS.

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Examiner Tips and Tricks

- The square roots will be negatives of each other when written in cartesian form, and the two square roots will be π radians apart when written in polar form. These two facts can help you find the roots quicker and/or check your answers.
- If your calculator is able to work with complex numbers, you should also square the square-roots you found to check that you get the original number.

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